



SCHOOL OF COMPUTER TECHNOLOGY

AASD 4008 Big Data Tools and Technologies

What is a OLAP Cube



Online analytical processing (OLAP) cubes are used on the existing data warehouse infrastructure to provide self-service business intelligence capabilities to end users.

An OLAP cube is a data structure that overcomes the limitations of relational databases by providing rapid analysis of data. Cubes can display and sum large amounts of data while also providing users with searchable access to any data points. This way, the data can be rolled up, sliced, and diced as needed to handle the widest variety of questions that are relevant to a user's area of interest.

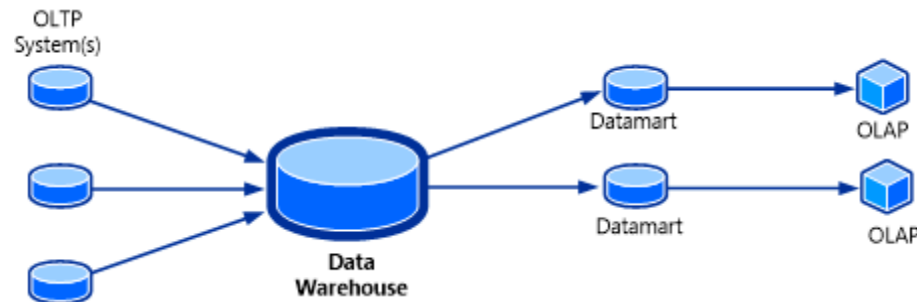
What is a OLAP Cube - Continued



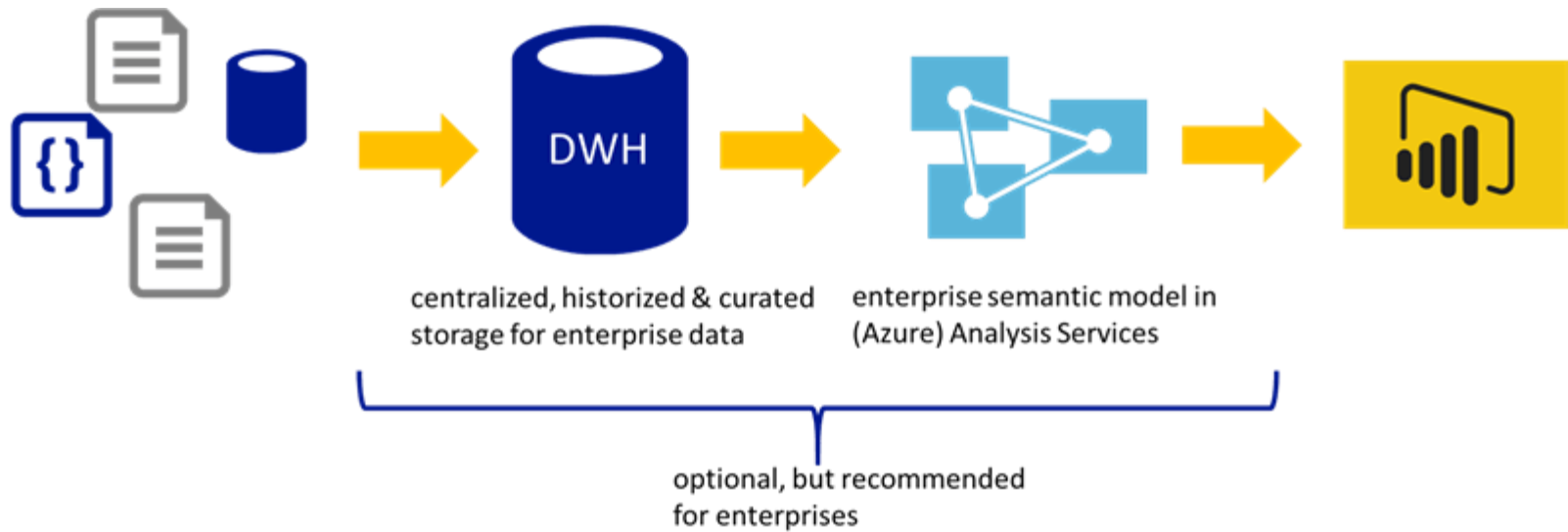
Software vendors or information technology (IT) developers with a working knowledge of OLAP cubes can create management packs to define their own extensible and customizable OLAP cubes that are built on the data warehouse infrastructure. These cubes are stored in SQL Server Analysis Services (SSAS). Self-service business intelligence tools such as Excel and SQL Server Reporting Services (SSRS) or Power BI /PowerPivot can target these cubes in SSAS, and you can use them to analyze the data from multiple perspectives.

What is a OLAP Cube - Continued

OLAP cubes can be considered as the final piece of the puzzle for a data warehousing solution. An OLAP cube, also known as multidimensional cube or hypercube, is a data structure in SQL Server Analysis Services (SSAS) that is built, using OLAP databases, to allow near-instantaneous analysis of data. The topology of this system is shown in the following illustration.



Reference architecture



Create Sql Server Analysis services instance – tabular on Azure



Microsoft Azure Search resources, services, and docs (G+/)

Home > Analysis Services >

Analysis Services

Analysis Services

Please fix the errors on this page before continuing.

Server name * ⓘ
nizaralitabularmodel ✓

Subscription *
paidsubscription ▼

Resource group *
alinizartempv2 ▼
[Create new](#)

Location *
East US ▼

Pricing tier (View full pricing details) *
D1 (20 Query Processing Units) ▼

Administrator (Select) * ⓘ
nizar.mithani@gmail.com#EXT#@nizarmithanigmail.onmicrosoft.com ✓

Backup Storage Settings
[Backup Storage: Not configured](#)

Storage key expiration
Never ▼

Validating... ⓘ Automation options

Step 1 – New Project



Create a new project

Recent project templates

 Integration Services Project

×

[Clear all](#)

All languages

All platforms

All project types



Analysis Services Multidimensional and Data Mining Project
An **Analysis** Services project for creating multidimensional and data mining models.



Analysis Services Multidimensional and Data Mining Project
An **Analysis** Services project for creating multidimensional and data mining models.



Analysis Services Tabular Project New
An **Analysis** Services project for creating tabular models.



Import from Server (Multidimensional and Data Mining)
Creates a multidimensional and data mining project by extracting the metadata from an existing multidimensional and data mining database on an **Analysis** Services server.

Step 1.5 – configure project



Configure your new project

Analysis Services Tabular Project

Project name

cubeprocessing_tabular_azureSSAS

Location

C:\Users\azureuser\Source\Repos

...

Solution

Create new solution

Solution name ⓘ

cubeprocessing_tabular_azureSSAS

☐ Place solution and project in the same directory

Framework

.NET Framework 4.7.2

Step 2 – Point to SSAS Azure



Tabular model designer

Select an Analysis Services instance to use while authoring projects.

☐ Integrated workspace

Integrated workspace eliminates the need to provide an explicit AS server instance.

☒ Workspace server

asazure://eastus.asazure.windows.net/nizaralitabulardmodel

Test Connection

In order to create a new Tabular model, you must select a compatibility level. The compatibility level must be compatible with the Analysis Services server version you want to deploy to.

Compatibility level:

SQL Server 2016 RTM (1200)

[Click here for more information about compatibility level.](#)

☐ Do not show this message again.

OK Cancel

Step 3 – Impersonation

Table Import Wizard

Impersonation Information
Specify the credentials used by the Analysis Services server to connect to the data source when importing and processing data.

☐ Specific Windows user name and password
Connects to the data source using the credentials of the user named below.

User Name:

Password:

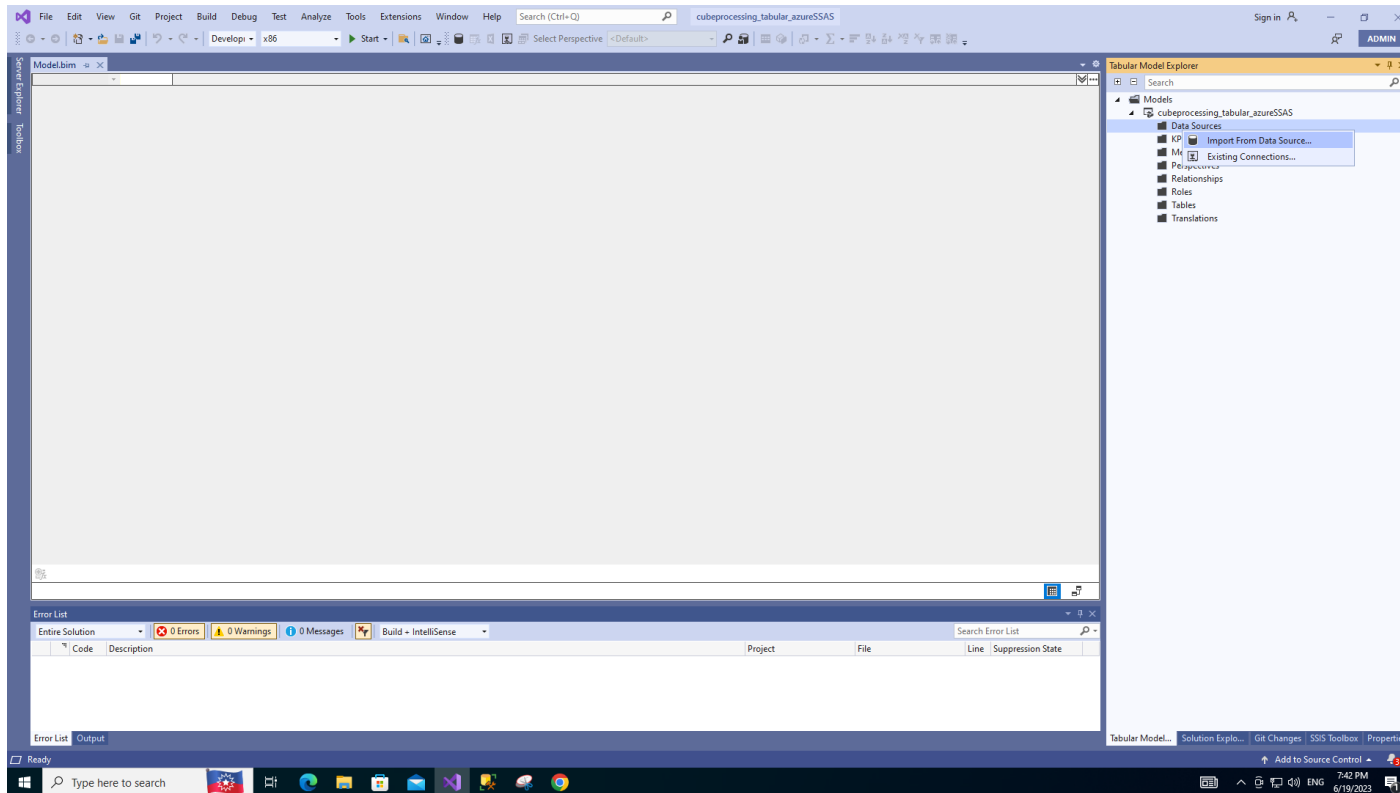
☒ Service Account
Connects to the data source using the credentials of the user running the Analysis Service server.

☐ Current User
Use the credentials of the current user.

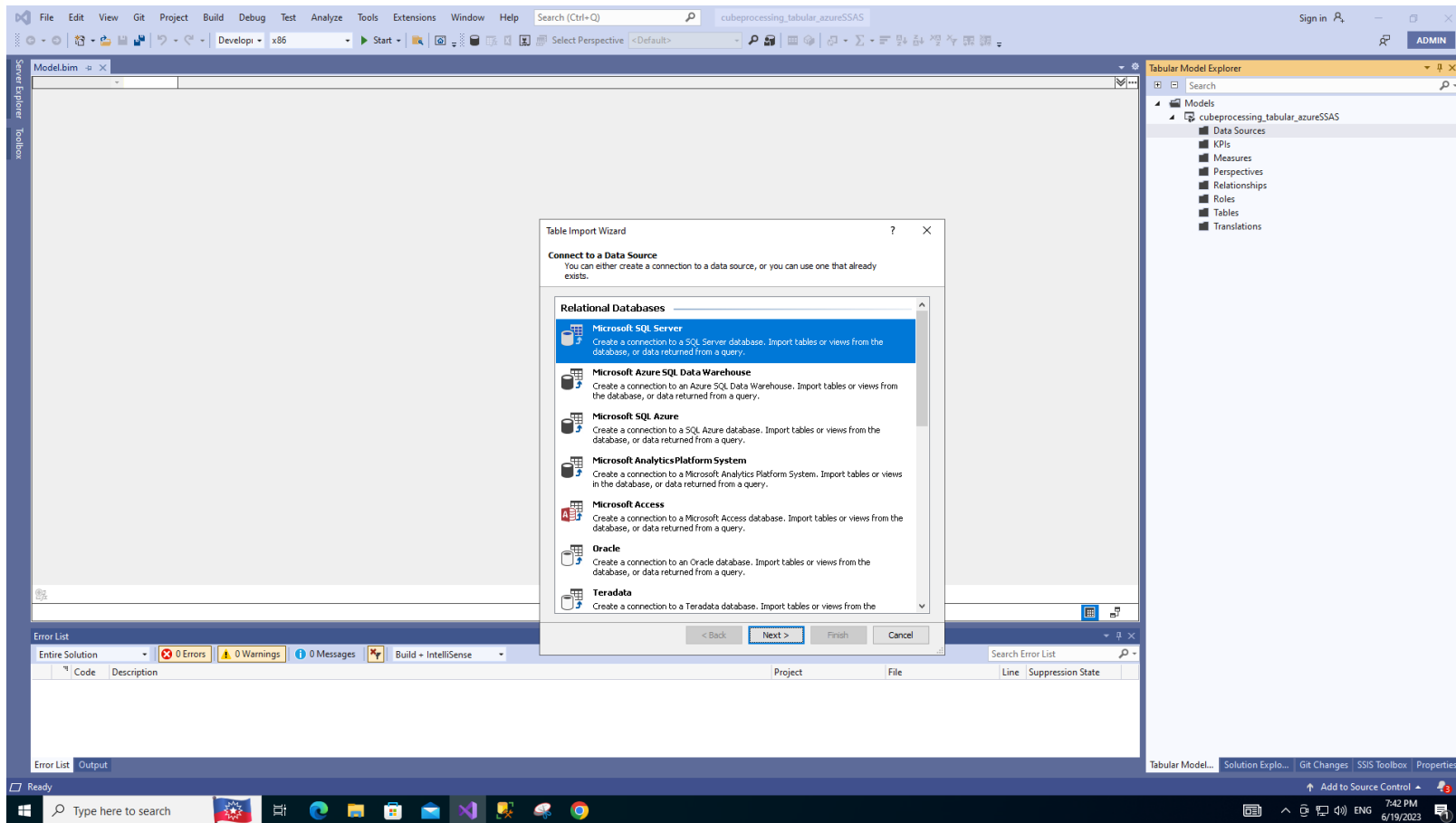
☐ Unattended Account
Connects to the data source using a low privilege account.

< Back Next > Finish Cancel

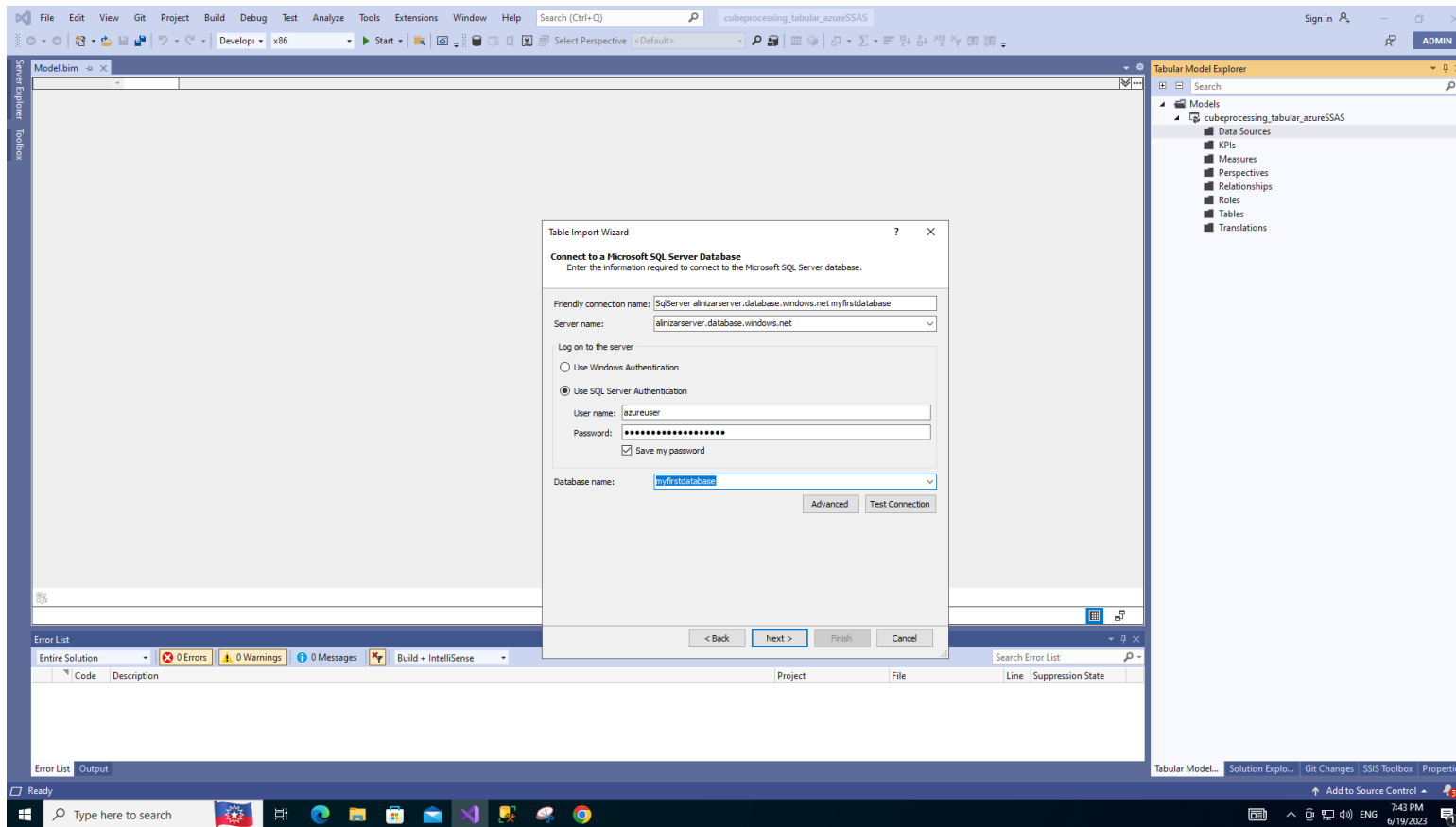
Step 4 – Import Data



Step 5 – Point to Azure SQL



Step 6 – Provide UserName/Password



Step 7 – Import Selected Data



Table Import Wizard

Select Tables and Views
Select the tables and views that you want to import data from.

Server: alnizarserver.database.windows.net
Database: myfirstdatabase

Tables and Views:

☐	Source Table	Schema	Friendly Name	Filter Details
☐	BuildVersion	dbo		
☐	ErrorLog	dbo		
☒	Address	Sales.T	Address	
☒	Customer	Sales.T	Customer	
☒	CustomerAddress	Sales.T	CustomerAddress	
☒	Product	Sales.T	Product	
☒	ProductCategory	Sales.T	ProductCategory	
☒	ProductDescription	Sales.T	ProductDescription	
☒	ProductModel	Sales.T	ProductModel	
☒	ProductModelProductDescrip...	Sales.T	ProductModelProductDescrip...	
☒	SalesOrderDetail	Sales.T	SalesOrderDetail	
☒	SalesOrderHeader	Sales.T	SalesOrderHeader	
☐	vGetAllCategories	Sales.T		
☐	vProductAndDescription	Sales.T		
☒	vProductModelCatalogDescri...	Sales.T		

< Back Next > Finish Cancel

Tabular Model Explorer

- Models
 - cubeprocessing_tabular_azureSSAS
 - Data Sources
 - KPIs
 - Measures
 - Perspectives
 - Relationships
 - Roles
 - Tables
 - Translations

Error List

Entire Solution 0 Errors 0 Warnings 0 Messages Build + IntelliSense

Search Error List

Line Suppression State

Ready

Type here to search

7:44 PM 6/19/2023

Tutorial



Students can also the following example from Microsoft for Furter reading

<https://docs.microsoft.com/en-us/analysis-services/tutorial-tabular-1400/as-lesson-1-create-a-new-tabular-model-project?view=asallproducts-allversions>